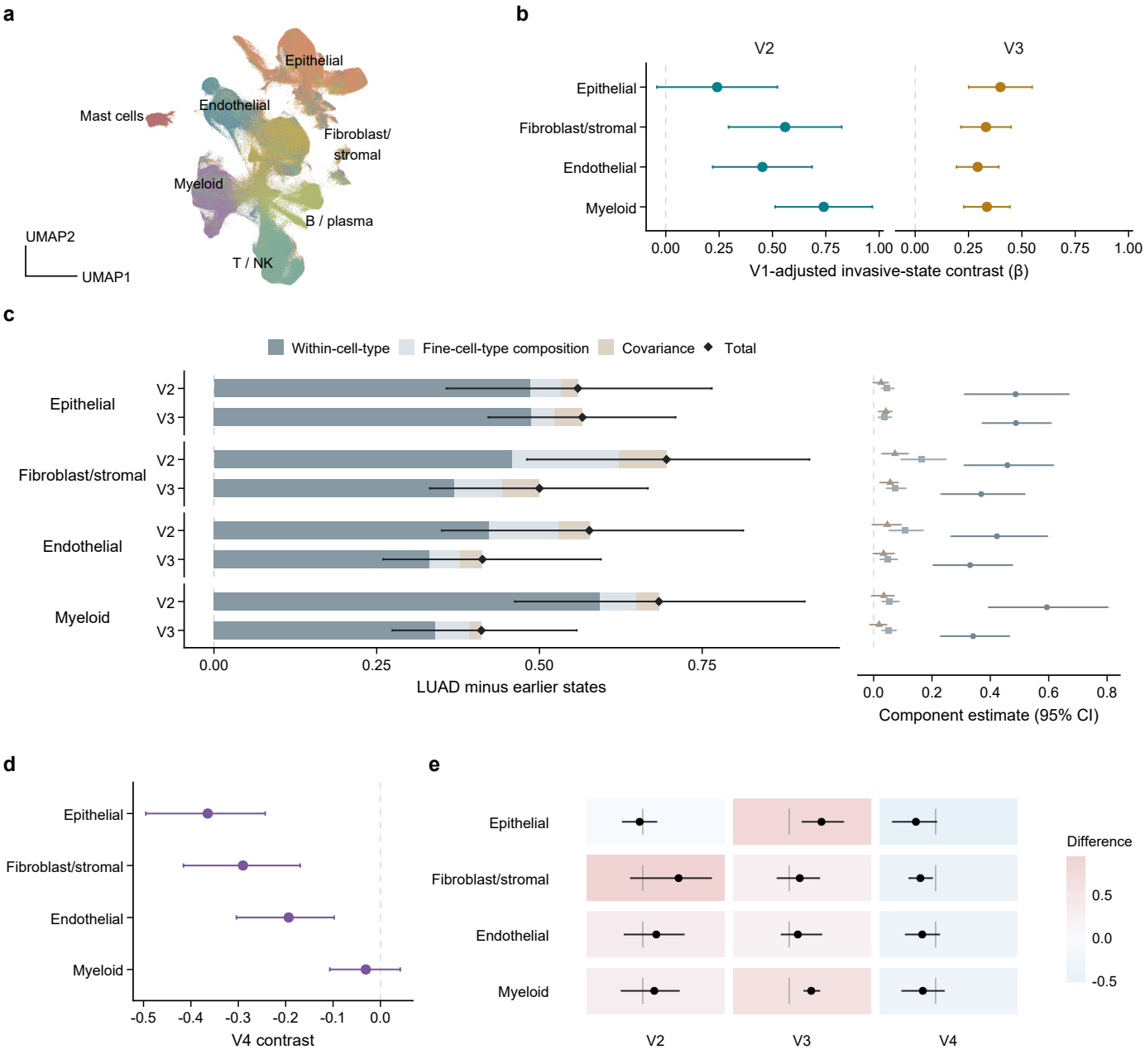

a
Epithelial
Endothelial
Mast cells
Fibroblast/
stromal
Myeloid
B / plasma
T / NK
UMAP2
UMAP1
b
V2
V3
Epithelial
Fibroblast/stromal
Endothelial
Myeloid
0.00
0.25
0.50
0.75
1.00
V1-adjusted invasive-state contrast (β)
c
Within-cell-type
Fine-cell-type composition
Covariance
Total
Epithelial
Fibroblast/stromal
Endothelial
Myeloid
V2
V3
0.00
0.25
0.50
0.75
LUAD minus earlier states
0.0
0.2
0.4
0.6
0.8
Component estimate (95% CI)
d
Epithelial
Fibroblast/stromal
Endothelial
Myeloid
-0.5
-0.4
-0.3
-0.2
-0.1
0.0
V4 contrast
e
Epithelial
Fibroblast/stromal
Endothelial
Myeloid
V2
V3
V4
Difference
0.5
0.0
-0.5